THE NATIONAL INSTITUTE OF ENGINEERING

South Campus, Mysuru – 570008

Department of Electronics and Communication Engineering

VIRTUAL LAB REPORT

Course: Linear Integrated Circuits (BEC456B)

Submitted By

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Section: C

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<u>Experiment Title:</u>

To study and measure the characteristics and parameters of an Op-amp (IC-741):

(a) Input bias current at inverting terminal

(b) Input bias current at non-inverting terminal

(c) Input offset current

(d) Input offset voltage

(e) Slew rate

1. Aim

To study the practical characteristics of the IC-741 Operational Amplifier and to experimentally measure the input bias currents (I_B1, I_B2), input offset current (I_io), input offset voltage (V_io), and the Slew Rate (SR).

2. Introduction to Operational Amplifiers

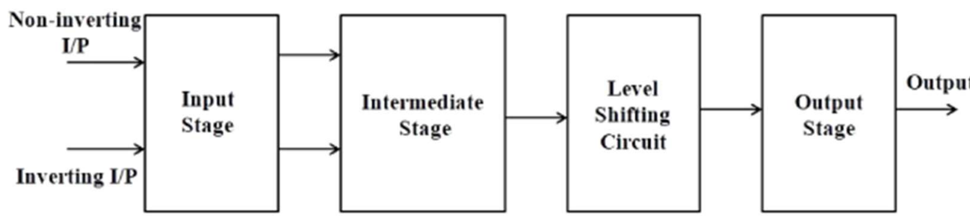
An **Operational Amplifier (Op-Amp)** is a high-gain, direct-coupled integrated circuit. Think of it as a "math machine" in a chip; it was originally designed to perform addition, integration, and subtraction in analog computers.

Block Diagram Analysis

An Op-Amp consists of four main stages that work together to create its massive gain:

1. **Input Stage:** A dual-input, balanced-output differential amplifier. This stage defines the input resistance.
2. **Intermediate Stage:** Provides additional voltage gain.

3. **Level Shifter:** Since the stages are direct-coupled, the DC level tends to rise. This stage shifts the DC level back down to zero volts.
4. **Output Stage:** Usually a push-pull complementary amplifier. This allows the Op-Amp to deliver significant current to a load with very low output resistance.



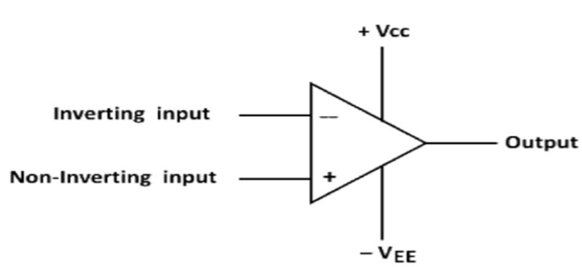
3. Schematic Symbol and Pins (IC-741)

The Op-Amp is represented by a triangle with two inputs and one output.

- **Inverting Input (-):** A signal here appears at the output with a 180° phase shift.
- **Non-Inverting Input (+):** A signal here appears at the output in the same phase.

IC-741 Pin Configuration

- **Pin 2 & 3:** Inverting and Non-inverting inputs.
- **Pin 4 & 7:** Negative and Positive power supplies (typically pm15V).
- **Pin 6:** The Output.
- **Pins 1 & 5:** Offset Null (used to "zero out" the output manually using a potentiometer).



4. Characteristics of an ideal op-amp:

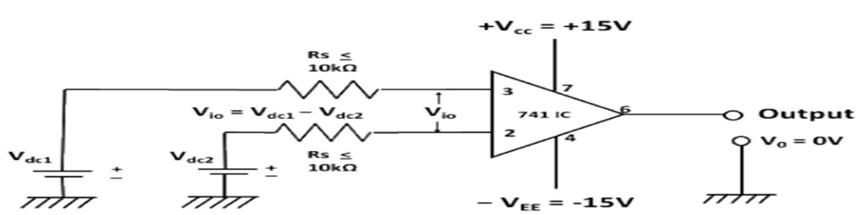
An ideal op-amp would exhibit the following electrical characteristics:

• Infinite voltage gain A.
• Infinite input resistance Ri, so that almost any signal source can drive it and there is no loading of the preceding stage.
• Zero output resistance Ro, so that the output can drive an infinite number of other devices.
• Zero output voltage when input voltage is zero.
• Infinite bandwidth, so that any frequency signal from 0 to ∞ Hz can be amplified without attenuation.
• Infinite common-mode rejection ratio (CMRR), so that the output common mode noise voltage is zero.
• Infinite slew rate, so that output voltage changes occur simultaneously with input voltage changes.

Parameters of an op-amp

Input offset voltage

Input offset voltage is the voltage that must be applied between the two input terminals of an op-amp to null the output as shown in Fig.3. In figure V_{dc1} and V_{dc2} are dc voltages and R_s represents the source resistance. We denote input offset voltage by V_{io}. This voltage V_{io} could be positive or negative. For a 741C, the maximum value of V_{io} is 6mV dc. The smaller the value of V_{io}, the better the input terminals are matched. For instance, the 741C precision op-amps has maximum V_{io}= 150μV.

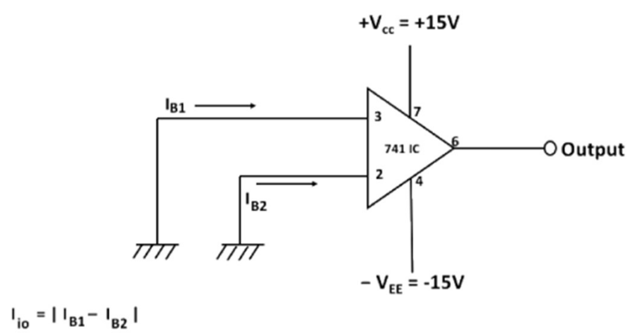


Input offset current

The algebraic difference between the currents into the inverting and non-inverting terminals is known as input offset current I_{io}. In the form of an equation,

$I_{io} = |I_{B1} - I_{B2}|$

where I_{B1} is the current into the non-inverting input and I_{B2} is the current into the inverting input. The input offset current for the 741C is maximum 200nA . As the matching between two input terminals is improved, the difference between $I_{B1}$and I_{B2} becomes smaller; i.e. the I_{io} value decreases further. For instance, the precision op-amp 741C has a maximum value of lio equal to 6nA, a dramatic improvement over older technology.



Input bias current

Input bias current I_B, is the average of the currents that flow into the inverting and non-inverting input terminals of the op-amp. In the form of an equation,

$I_B = (I_{B1} + I_{B2})/2$

I_B = 500nA (maximum) for 741C, whereas for the precision 741C, it is ±7nA. The two input currents I_{B1} and I_{B2} are actually the base currents of the first differential amplifier stage.

Common mode rejection ratio:

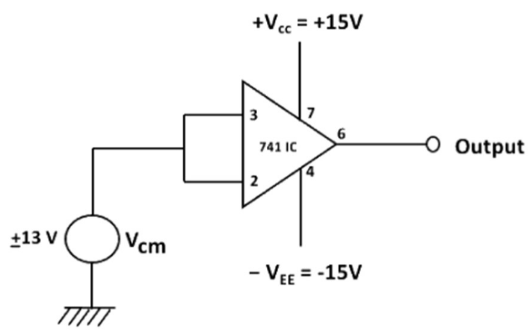
The common-mode rejection ratio (CMRR) is defined as the ratio of differential voltage gain A_d to the common mode voltage gain A_{cm} i.e,

CMRR $=A_d/ A_{cm}$

The differential voltage gain A_d is same as the large-signal voltage gain A, however, the common-mode voltage gain can be determined from the circuit of as

$A_{cm} = V_{ocm} / V_{cm}$

where -
V_{ocm} = output common mode voltage
V_{cm} = input common mode voltage
A_{cm} = common mode voltage gain



Generally, the A_{cm} is very small and Ad = A is very large; therefore, the CMRR is very large. Being a very large value, the CMRR is most often expressed in decibels(dB). For the 741C,CMRR is typically 90dB. This value of CMRR is determined under the test condition that the input source resistance R_s<10KΩ. In R_s is assumed to be zero because most of the practical voltage sources have negligible source resistances.

The higher the value of CMRR, the better is the matching between the two input terminals and the smaller is the output common mode voltage. For the 741C precision op-amp, CMRR =120dB. This means that the precision 741C has a better ability to reject common-mode voltages, such as electrical noise, than the 741C and is preferred in noise environments.

Slew rate

Slew Rate (SR) is defined as the maximum rate of change of output voltage and is expressed in volts per micro-seconds

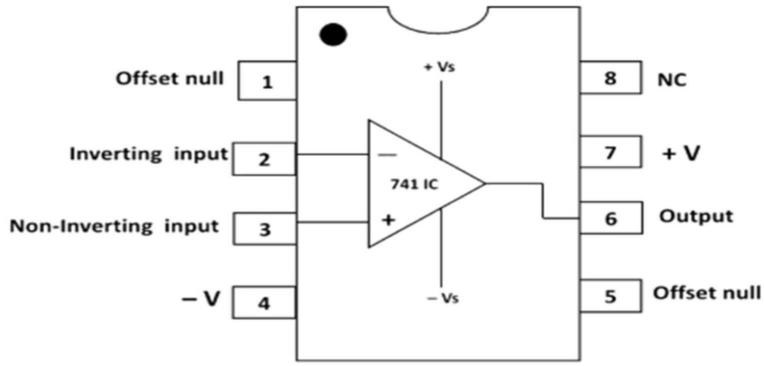
$SR= (dV_o /dt)_{max}$ V/μs,

Slew rate indicates how rapidly the output of an op-amp can change in response to changes in the input frequency. The slew rate changes with change in voltage gain and is normally specified at unity (+1) gain. The slew rate of an op-amp is fixed; therefore, if the slope requirements of the output signal are greater than the slew rate, distortion occurs. The slew rate is one of the important factors in selecting the op-amp for ac applications, particularly at relatively high frequencies.

One of the drawbacks of the 741C is its low slew rate (0.5V/μs), which limits its use in relatively high frequency applications, especially in oscillators, comparators and filters. The newer op-amps: LF351, μAF771 and MC 34001 which are direct replacements for 741, have a slew rate of 13V/μs. In high-speed op-amps, especially, the slew rate is significantly improved. For instance, the LM318 has a slew rate of 70V/μs.

Op-amp IC-741 pin configuration

The pin configuration of IC-741 general purpose operational amplifier is shown in The reference point of the chip is the notch at the top. The pins are numbered in a counter clockwise direction, starting at the notch.



➢ **Pin 1** and **Pin 5** are offset null pins. Offset null is a calibration feature of the op-amp. The op-amp is so sensitive to the input voltage that at times, the output will generate a signal even when there is no intentional input. To avoid this condition for certain applications, offset null pins are provided. They are usually connected to a variable resistance such as a potentiometer. The potentiometer can be adjusted to produce a zero-output voltage from Pin 6.

➢ **Pin 2** and **Pin 3** are two input pins. Both inputs, pin 2 and pin 3, generate an output at Pin 6, but they generate opposite polarities. Pin 2 is the inverting input. Any signal applied to Pin 2 generates a signal of opposite polarity at the output. The non-inverting input, Pin 3, generates an output of the same polarity at Pin 6.

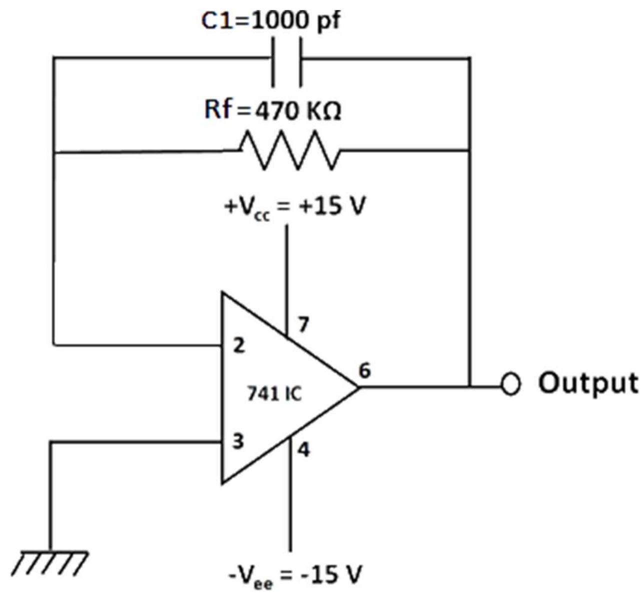
➢ Power to the op-amp is provided through **Pin 4** and **Pin 7**. Pin 4 is either connected to ground or a negative voltage value from 3 to 18 volts. Pin 7 is connected to the positive voltage of the power supply.

➢ The output of the amplifier is at **Pin 6**.

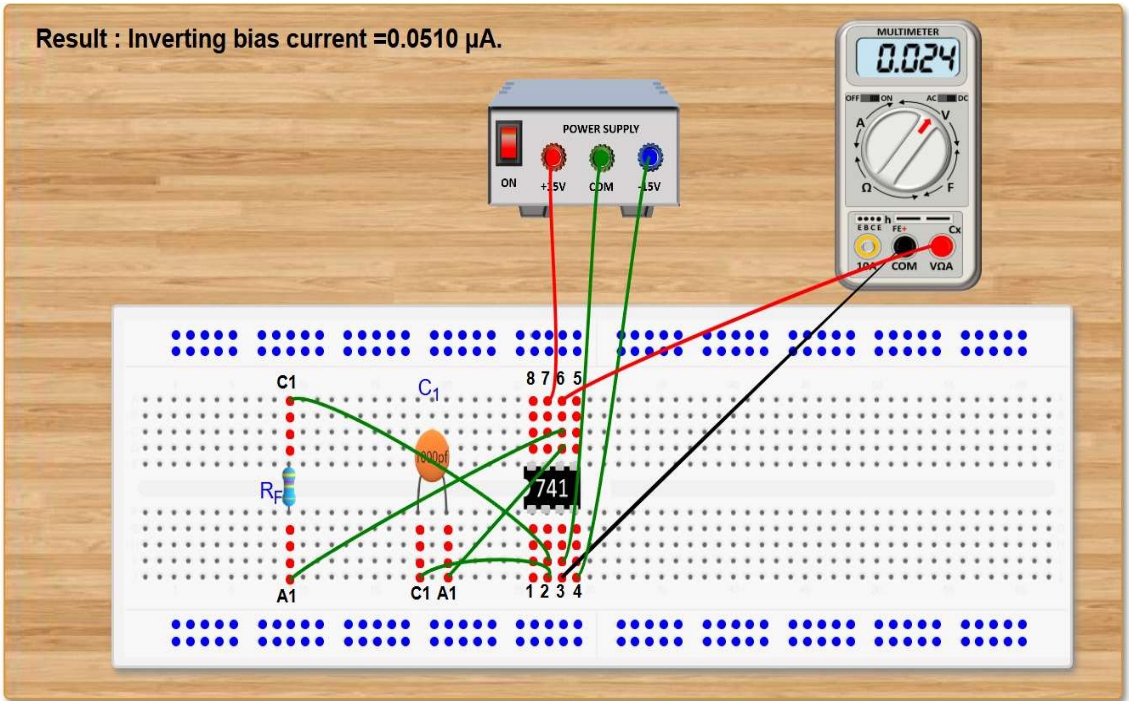
➢ **Pin 8** is not used. 'NC' stands for no connection.

A. Measurement of Input Bias Current I_B

a) Measurement of inverting bias current IB1



S. No.	Source	Target
1	IC741_ pin 07	Supply VCC
2	IC741_ pin 03	Supply GND
3	Multimeter _GND	Supply GND
4	IC741_ pin 04	Supply VEE
5	IC741 _ pin 02	Resistance (R_f) _ terminal C1
6	IC741_ pin 06	Resistance (R_f) _ terminal A1
7	IC741_ pin 02	Capacitance (C_1) _ terminal C1
8	IC741_ pin 06	Capacitance (C_1) _ terminal A1
9	Multimeter_VCC	IC741_ pin 06



Measurement of Inverting Bias Current (I_B2)

In this specific Virtual Lab setup, you are measuring the DC current that flows into the **Inverting Terminal (Pin 2)** of the Op-Amp.

1. Circuit Analysis from the Image

- **Non-Inverting Terminal (Pin 3):** This is connected directly to the Common Ground (COM).
- **Inverting Terminal (Pin 2):** A feedback resistor R_F is connected between the output (Pin 6) and the inverting input (Pin 2).
- **Input Stage:** Since the non-inverting terminal is grounded, the current required to bias the inverting terminal must be supplied through the feedback loop.
- **Result Displayed:** The simulation shows a result of **0.0510 µA**.

2. The Formula

The bias current is derived from Ohm's Law by measuring the output voltage created by the current flowing through a known resistor.

When the non-inverting terminal is grounded, the output voltage V_o is the product of the inverting bias current and the feedback resistor:

V_o = I_B2 times R_F

To find the **Inverting Bias Current (I_B2)**, we rearrange the formula:

I_B2 = V_o/R_F

Where:

- **I_B2**: Inverting input bias current (typically in nA or μA).
- **V_o**: Output voltage measured by the multimeter.
- **R_F**: Feedback resistor used in the circuit.

Resistance Rf (Ω)	**Measured Output Voltage Vo (V)**	**Calculated Inverting Bias Current IB− (μA)**
470k	0.024	0.0510

- V_o = 0.024 V
- R_f = 470kOmega = 470,000Omega

Calculation:

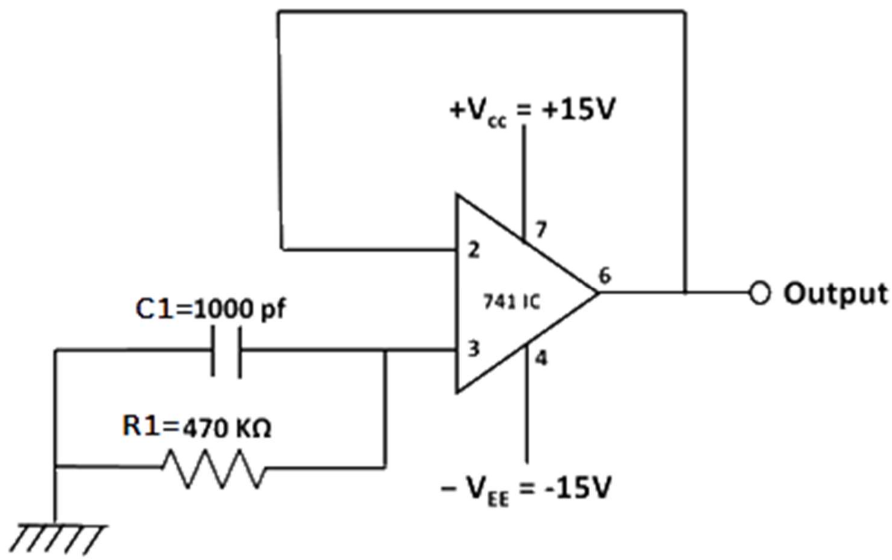
I_B2 = 0.024/470,000

=> 0.00000005106 A

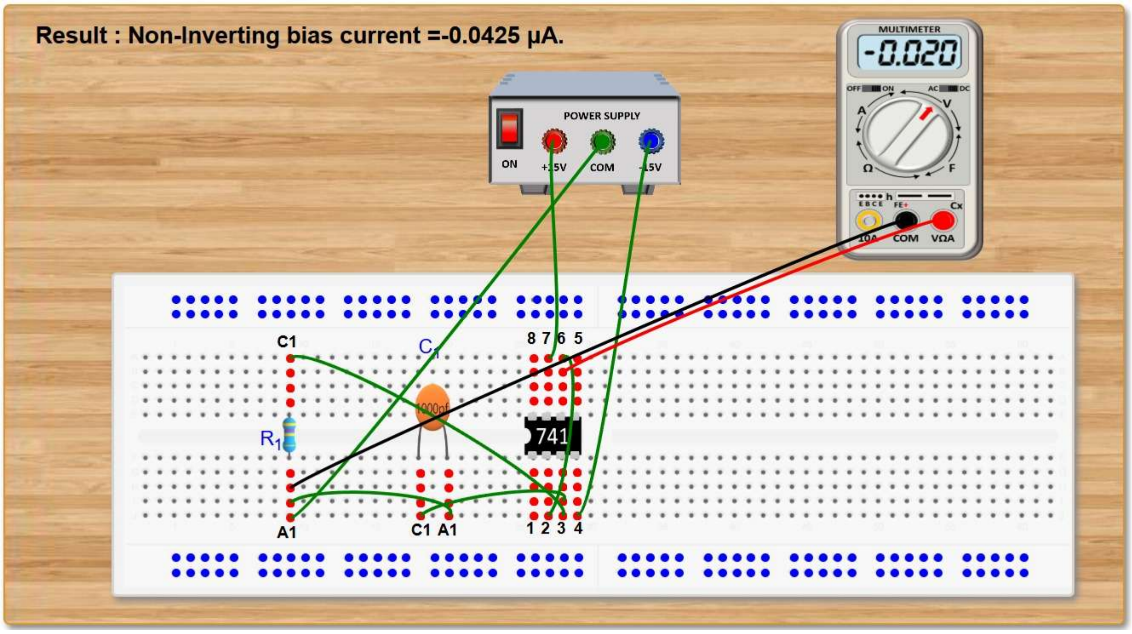
I_B2 = 0.0510 uA

Result: The **Inverting Bias Current (I_B2)** for the given Op-Amp (IC-741) is measured to be **0.0510 uA**.

b) Measurement of non-inverting bias current I_{B2}



S. No.	Source	Target
1	IC741_ pin 07	Supply VCC
2	Multimeter_GND	Supply GND
3	IC741_ pin 06	IC741_ pin 02
4	IC741_ pin 04	Supply VEE
5	IC741_ pin 03	Resistance (R_1) _terminal C1
6	Supply GND	Resistance (R_1) _terminal A1
7	IC741_ pin 03	Capacitance (C_1) _terminal C1
8	Supply GND	Capacitance (C_1) _terminal A1
9	Multimeter_VCC	IC741_ pin 06



Measurement of Non-Inverting Bias Current (I_B1)

In this circuit, you are measuring the DC current flowing into the **Non-Inverting Terminal (Pin 3)** of the Op-Amp.

1. Circuit Analysis from the Image

- **Inverting Terminal (Pin 2):** This is connected directly to the Common Ground (COM).
- **Non-Inverting Terminal (Pin 3):** A resistor R_1 is connected between this terminal and the ground.
- **Output Voltage (V_o):** The multimeter shows a reading of **-0.020 V**.
- **Resulting Current:** The simulation calculates a non-inverting bias current of **-0.0425 µA**.

2. The Formula

The non-inverting bias current is calculated by measuring the output voltage produced when the current flows through a known resistance at the non-inverting input. For this configuration, the formula is:

I_B1 = V_o/R_1

Where:

- **I_B1**: Non-Inverting input bias current.
- **V_o**: Output voltage measured by the multimeter.
- **R_1**: The resistor connected to the non-inverting terminal.

Resistance R1 (Ω)	Measured Output Voltage Vo (V)	Calculated Non-Inverting Bias Current IB1 (μA)
470k	-0.020	-0.0425

Calculation

Using the values from your simulation:

- V_o = -0.020 V
- R_1 = 470kOmega = 470,000Omega

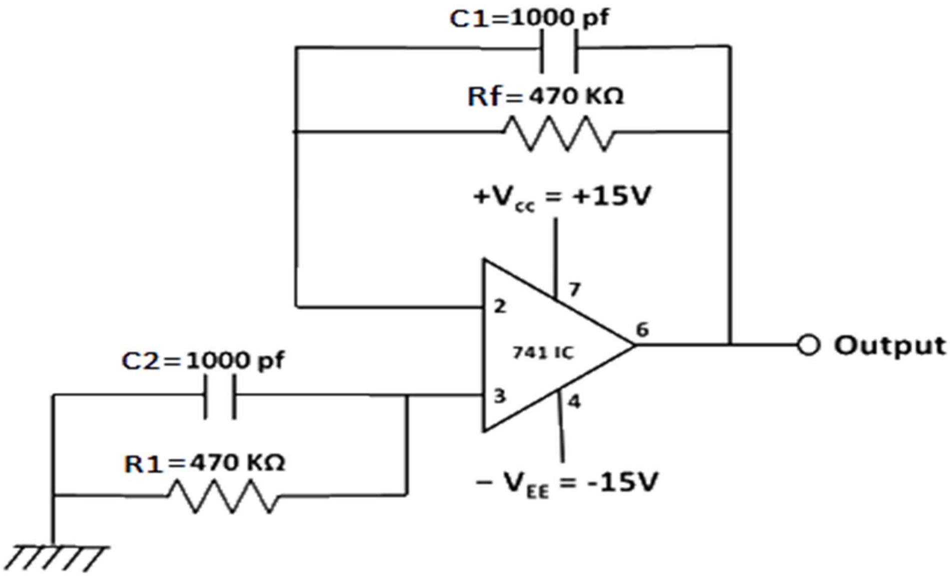
I_B1 = -0.020/470,000

=> -0.00000004255 A

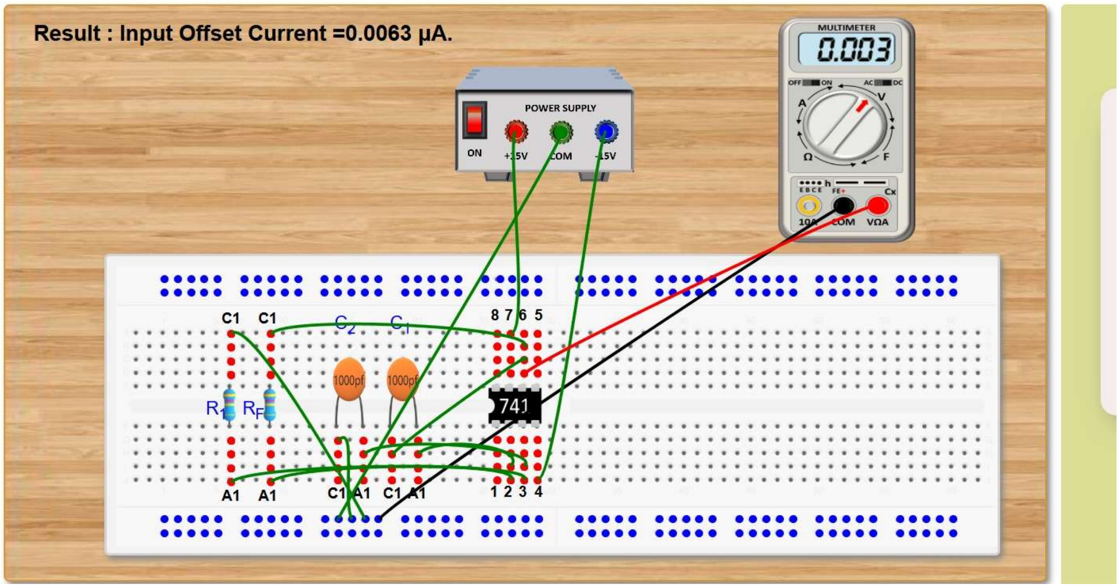
I_B1 =-0.0425 uA

Result: The **Non-Inverting Bias Current (I_B1)** for this IC-741 Op-Amp is **-0.0425 muA**.

B. Measurement of input offset current I_{io}



S. No.	Source	Target
1	IC741_ pin 07	Supply VCC
2	Multimeter_GND	Supply GND
3	IC741_ pin 04	Supply VEE
4	Resistance (R_f)_terminal A1	IC741_ pin 02
5	Resistance (R_f)_terminal C1	IC741_ pin 06
6	Capacitance (C_1) _terminal A1	IC741_ pin 02
7	Capacitance (C_1) _terminal C1	IC741_ pin 06
8	Resistance (R_1) _ terminal A1	IC741_ pin 03
9	Resistance (R_1) _ terminal C1	Supply GND
10	Capacitance (C_2) _terminal A1	IC741_ pin 03
11	Capacitance (C_2) _terminal C1	Supply GND
12	Multimeter_VCC	IC741_ pin 06



1. Definition and Formula

As stated in your theory, the **Input Offset Current (I_io)** is the algebraic difference between the currents flowing into the inverting and non-inverting terminals.

The formula used to calculate this in your lab is:

I_io = |I_B1 - I_B2|

Where:

- **I_B1**: Bias current at the non-inverting terminal.
- **I_B2**: Bias current at the inverting terminal.

In a practical measurement using the output voltage (V_o) from your third image, the relationship is:

I_io = V_o/R_f

2. Circuit Analysis from the Image

In this setup, resistors are connected to **both** input terminals (Pins 2 and 3). This configuration allows the Op-Amp to amplify the mismatch between the two input currents.

- **Output Voltage (V_o)**: The multimeter in the third image shows **0.003 V**.
- **Feedback Resistor (R_f)**: Based on the previous steps, this is **470 kOmega**.
- **Result**: The simulation calculates the offset current as **0.0063 μA**.

3. Observation Table

Parameter	Resistance Rf (Ω)	Measured Output Voltage Vo (V)	Calculated Input Offset Current Iio (μA)
Input Offset Current	470k	0.003	0.0063

4. Calculation Verification

Using the formula and the values

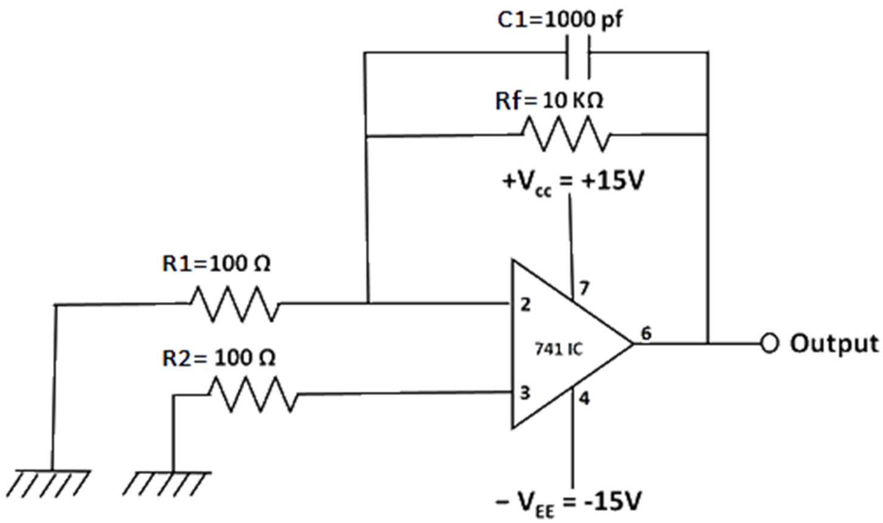
I_io =0.0063 muA

Alternatively, using the individual bias currents measured in previous parts:

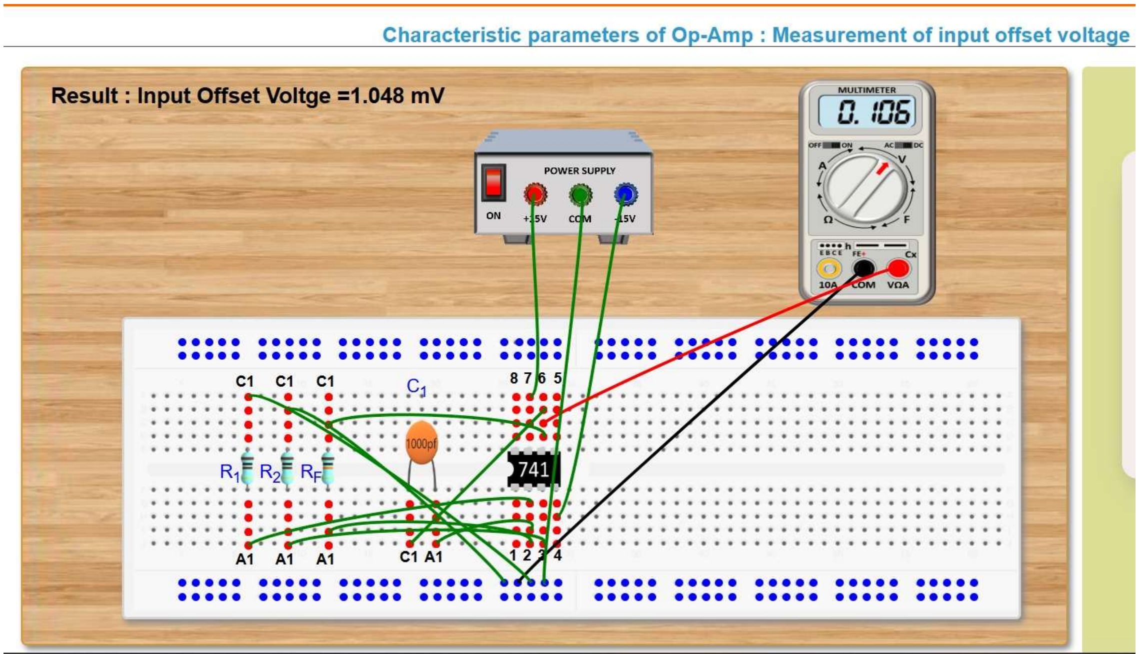
- I_B2 (Inverting) = 0.0510 μA
- I_B1 (Non-Inverting) = 0.0425 μA (magnitude)
- I_io = |0.0425 - 0.0510| = 0.0085 muA
- *Note: The small discrepancy between the subtraction and the direct V_o measurement is common in simulations due to internal component tolerances modeled in the Virtual Lab.*

Result : The measured **Input Offset Current (I_io)** for the IC-741 is **0.0063 μA**

C. Measurement of input offset voltage V_{io}



S. No.	Source	Target
1	IC741_ pin 07	Supply VCC
2	Multimeter_GND	Supply GND
3	IC741_ pin 04	Supply VEE
4	Resistance (R_f)_terminal A1	IC741_ pin 02
5	Resistance (R_f)_terminal C1	IC741_ pin 06
6	Capacitance (C_1) _terminal A1	IC741_ pin 02
7	Capacitance (C_1) _terminal C1	IC741_ pin 06
8	Resistance (R_2) _terminal A1	IC741_ pin 03
9	Resistance (R_2) _terminal C1	Supply GND
10	Resistance (R_1) _terminal A1	IC741_ pin 02
11	Resistance (R_1) _terminal C1	Supply GND
12	Multimeter_VCC	IC741_ pin 06



1. Theoretical Definition

In an ideal Op-Amp, if both input terminals are grounded (0 V), the output should be exactly 0 V. However, due to internal mismatches in the differential input stage (the transistors inside the chip), a small DC voltage exists at the output even with zero input. The **Input Offset Voltage (V_{io})** is defined as the voltage that must be applied between the two input terminals to force that output back to zero.

2. The Formula and Variables

To measure such a tiny voltage (usually in millivolts), we use a high-gain circuit to amplify the offset to a level that a standard multimeter can read. The formula used in your lab is:

$V_{io} = V_o -/ (I_{io} * R_f)1 + R_f/R_i$

- **V_o (Output Voltage):** This is the amplified "error" voltage measured at the output pin (0.106 V in your data).
- **R_f and R_i:** These resistors determine the circuit gain. Your values are 100kOmega and 1kOmega respectively.

- **(1 + R_f/R_i):** This is the **Closed-Loop Gain**. In your case, 1 + (100,000 / 1,000) = 101.
- **I_io * R_f:** This term accounts for the error contributed by the Input Offset Current. If the offset current is very small, this term becomes negligible, simplifying the formula to V_io = V_o / Gain.

3. Calculation Walkthrough

Using the specific values from your observation table:

1. **Identify the Gain:**

Gain = 1 + 100kOmega/1kOmega = 101

2. **Calculate V_io (ignoring I_io for the approximation):**

V_io = 1.049 mV

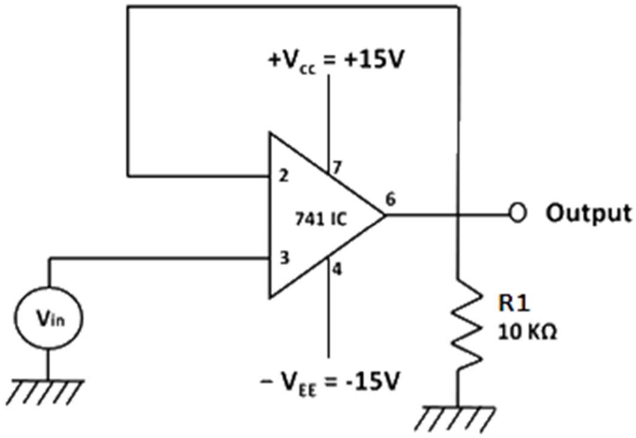
The result displayed in your simulation, **1.048 mV**, accounts for the tiny subtraction of the offset current (I_io), making it even more precise.

4. Observation Table

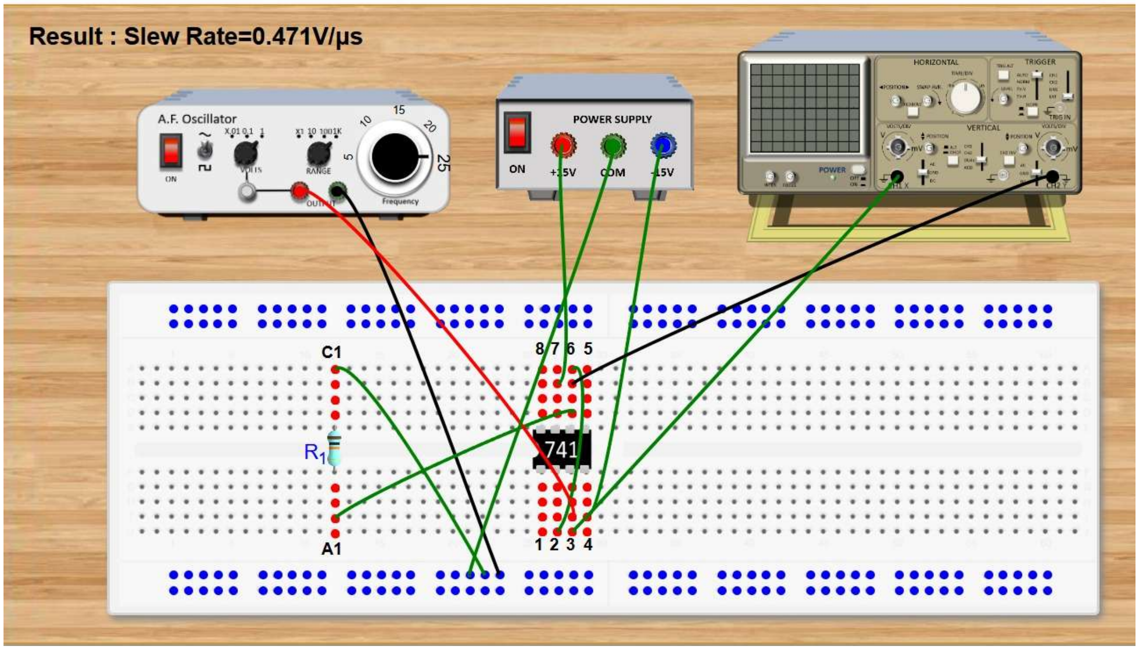
Resistance Ri	Resistance Rf	Output Voltage Vo	Calculated Vio
1kOmega	100kOmega	0.106 V	**1.048 mV**

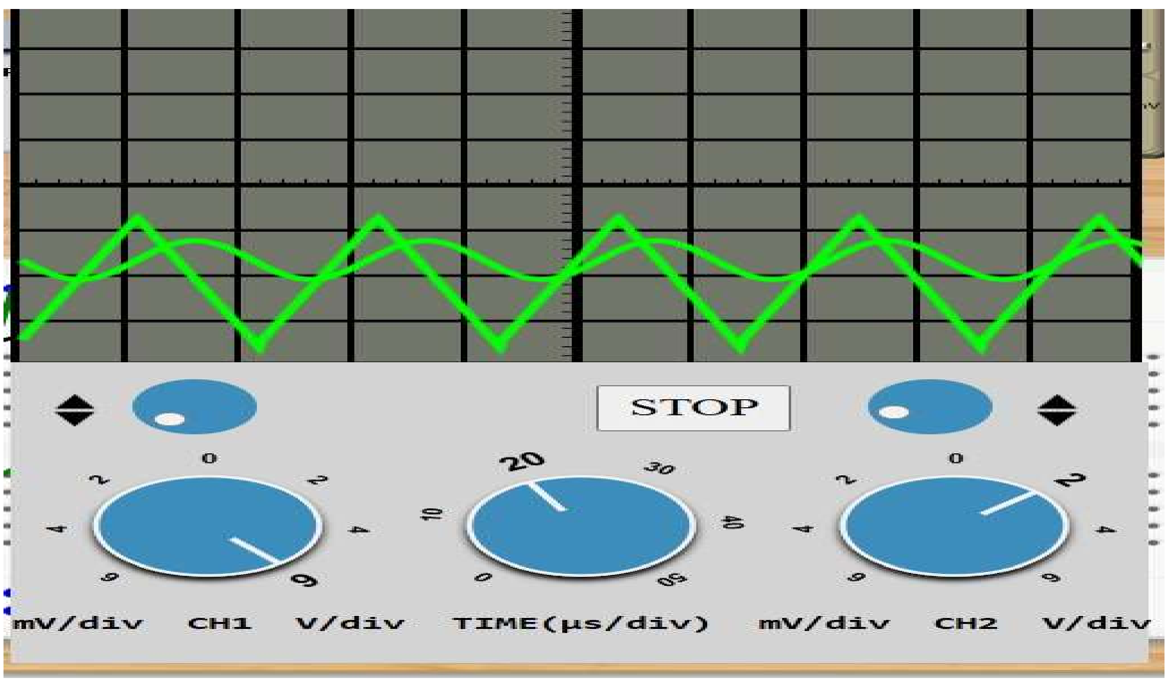
Result : For a standard **IC-741**, the maximum V_io is typically **6 mV**. Your measured value of **1.048 mV**

D. Measurement of slew rate S.R



S. No	Source	Target
1	IC741_ pin 07	Supply VCC
2	IC741_ pin 04	Supply VEE
3	Generator_terminal C	Supply GND
4	Resistance (R_1) _terminal A1	IC741_ pin 06
5	Resistance (R_1) _terminal C1	Supply GND
6	IC741_ pin 06	IC741_ pin 02
7	Generator_ terminal A	IC741_ pin 03
8	CRO_ terminal C	IC741_ pin 06
9	Generator_terminal A	CRO_terminal A


Result : Slew Rate=0.471V/µs
A.F. Oscillator
POWER SUPPLY
ON
+15V
COM
-15V
HORIZONTAL
TRIGGER
VERTICAL
C1
R1
741
A1
8 7 6 5
1 2 3 4


STOP
mV/div CH1 V/div
TIME(µs/div)
mV/div CH2 V/div

1. Slew Rate (SR) Formula for Sinusoidal Signals

The Slew Rate can be determined using the maximum frequency (f_max) at which an undistorted sine wave of a certain peak amplitude (V_m) can be obtained. The formula is:

SR = 2*pi *f_max *V_m/10^6   V/μs

- **2*pi* f_max* V_m**: This represents the maximum rate of change of a sine wave (the derivative at the zero-crossing).
- **10^6**: This factor converts the unit from Volts per second (V/s) to **Volts per microsecond (V/μs)**, which is the standard industry unit for Op-Amps.

2. Calculation Walkthrough

Based on your provided reference calculation, your simulation used the following parameters:

- **Maximum Frequency (f_max)**: 25 kHz.
- **Peak Output Voltage (V_m)**: 3 V.

Applying the values:

SR = 0.471 V/μs

3. Physical Interpretation

The Slew Rate indicates the "speed limit" of the Op-Amp's output stage.

- **Response Speed**: A value of **0.471 V/μs** means that for every one microsecond that passes, the output voltage can only climb or fall by a maximum of 0.471 Volts.
- **Distortion**: If you try to run a 25 kHz signal at an amplitude higher than 3 V, the Op-Amp will be unable to keep up with the steep slope of the sine wave. The output will begin to look "flattened" or triangular rather than a smooth curve.

4. Observation Table

Parameter	Input Frequency (f)	Peak Voltage (Vm)	Calculated Slew Rate	Typical IC-741 Value
Slew Rate	25 kHz	3 V	**0.471 V/μs**	0.5 V/μs

Result: The measured Slew Rate of **0.471 V/μs** is consistent with the standard performance of a general-purpose IC-741.

Conclusion

Through this Virtual Lab experiment, the practical characteristics of the IC-741 Op-Amp were successfully measured and verified against theoretical "ideal" values. While an ideal Op-Amp has zero offset and infinite speed, the practical IC-741 exhibits small but measurable errors such as input bias currents and a finite slew rate.

The experimental results closely align with the manufacturer's data sheets, confirming that the IC-741 is a reliable general-purpose amplifier for DC and low-frequency AC circuits, provided that the signal requirements do not exceed its internal physical limitations (specifically the Slew Rate).